

```
Processando config 1, seed 0, teste: BlockAlphabit
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L$  = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
```

```
Correction factor of the ChiSquare:
```

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 2
Value of the statistic : 5.95
p-value of test : 0.05
```

```
-----
CPU time used : 00:00:00.03
```

```
Generator state:
```

```
16777216 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L$  = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 4.17
p-value of test : 0.84

CPU time used : 00:00:00.03

Generator state:

33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 139.41
p-value of test : 0.23

CPU time used : 00:00:00.03

Generator state:

50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32666.45
p-value of test : 0.65

CPU time used : 00:00:00.03

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 8965.93
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305	
Chi-square statistic	:	3438.21	
p-value of test	:	eps	*****

CPU time used : 00:00:00.01

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	0.41	
p-value of test	:	0.34	

CPU time used : 00:00:00.01

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32	
ChiSquare statistic	:	751.46	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	31	
ChiSquare statistic	:	1734.40	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	32	
ChiSquare statistic	:	524.68	
p-value of test	:	eps	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	25	
ChiSquare statistic	:	71.63	
p-value of test	:	2.2e-6	*****

Test on the values of the Statistic C

Number of degrees of freedom	:	15	
ChiSquare statistic	:	37.07	
p-value of test	:	1.2e-3	

CPU time used : 00:00:00.07

Generator state:

134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	582.61	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	580.45	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	227.95	
p-value of test	:	3.4e-4	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	63.65	
p-value of test	:	0.25	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	39.07	
p-value of test	:	0.18	

CPU time used : 00:00:00.05

Generator state:
150994944 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.32
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):

(eps1 means a value < 1.0e-15):

	Test	p-value
5	HammingIndep, L = 16	eps
6	HammingIndep, L = 32	eps
8	RandomWalk1 H (L = 64)	eps
8	RandomWalk1 M (L = 64)	eps
8	RandomWalk1 J (L = 64)	eps
8	RandomWalk1 R (L = 64)	2.2e-6
9	RandomWalk1 H (L = 320)	eps
9	RandomWalk1 M (L = 320)	eps
9	RandomWalk1 J (L = 320)	3.4e-4

All other tests were passed

XXX
Starting Alhabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 0.57
p-value of test : 0.75

CPU time used : 00:00:00.03

Generator state:
167772160 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 17.91
p-value of test : 0.02

CPU time used : 00:00:00.01

Generator state:

184549376 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128

Value of the statistic : 148.38
p-value of test : 0.11

CPU time used : 00:00:00.01

Generator state:
201326592 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32772.27
p-value of test : 0.49

CPU time used : 00:00:00.03

Generator state:
218103808 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 177  
Chi-square statistic              : 168.04  
p-value of test                   : 0.67  
-----
```

```
-----  
CPU time used                     : 00:00:00.01  
-----
```

Generator state:

234881024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

```
-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0  
-----
```

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 305  
Chi-square statistic              : 322.22  
p-value of test                   : 0.24  
-----
```

```
-----  
CPU time used                     : 00:00:00.01  
-----
```

Generator state:

251658240 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingCorr test:

```
-----  
N = 1, n = 524288, r = 0, s = 32, L = 32  
-----
```

```
-----  
Normal statistic                  : -1.37  
p-value of test                   : 0.91  
-----
```

CPU time used : 00:00:00.01

Generator state:
268435456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	37.34
p-value of test	:	0.24

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	43.68
p-value of test	:	0.07

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	23.85
p-value of test	:	0.85

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	28.59
p-value of test	:	0.28

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	19.74
p-value of test	:	0.18

CPU time used : 00:00:00.07

Generator state:
285212672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	900.47	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	366.62	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	265.11	
p-value of test	:	3.4e-7	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	55.95	
p-value of test	:	0.51	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	34.71	
p-value of test	:	0.34	

CPU time used : 00:00:00.05

Generator state:
301989888 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.29
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	3.4e-7

All other tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	0.78
p-value of test	:	0.68

CPU time used : 00:00:00.01

Generator state:
318767104 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 3.78
p-value of test : 0.88

CPU time used : 00:00:00.01

Generator state:
335544320 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256

Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 118.50
p-value of test : 0.71

CPU time used : 00:00:00.00

Generator state:
352321536 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32815.75
p-value of test : 0.43

CPU time used : 00:00:00.03

Generator state:
369098752 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 164.28
p-value of test : 0.74

CPU time used : 00:00:00.01

Generator state:
385875968 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 6702.70
p-value of test : eps *****

CPU time used : 00:00:00.01

Generator state:
402653184 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.19
p-value of test : 0.43

CPU time used : 00:00:00.01

Generator state:

419430400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32
ChiSquare statistic : 2617.74
p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 31
ChiSquare statistic : 791.79
p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom : 32
ChiSquare statistic : 104.54
p-value of test : 1.3e-9 *****

Test on the values of the Statistic R

Number of degrees of freedom : 25
ChiSquare statistic : 47.76
p-value of test : 4.0e-3

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	17.82
p-value of test	:	0.27

CPU time used : 00:00:00.07

Generator state:
436207616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	927.55	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	470.07	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	352.03	
p-value of test	:	2.2e-16	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	72.49
p-value of test	:	0.08

Test on the values of the Statistic C

Number of degrees of freedom : 32
ChiSquare statistic : 45.12
p-value of test : 0.06

CPU time used : 00:00:00.05

Generator state:
452984576 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.26
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

	Test	p-value
6	HammingIndep, L = 32	eps
8	RandomWalk1 H (L = 64)	eps
8	RandomWalk1 M (L = 64)	eps
8	RandomWalk1 J (L = 64)	1.3e-9
9	RandomWalk1 H (L = 320)	eps
9	RandomWalk1 M (L = 320)	eps
9	RandomWalk1 J (L = 320)	2.2e-16

All other tests were passed

XX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216

Number of cells = $2^L =$ 4
Expected number per cell = 4194304
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

$\Delta = 1$, $\mu = 2.9802323e-08$, $\sigma = 1$

Test Results for $\Delta = 1.0000$

Number of degrees of freedom : 2
Value of the statistic : 1.50
p-value of test : 0.47

CPU time used : 00:00:00.01

Generator state:
469761792 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = $2^L =$ 16
Expected number per cell = 1048576
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

$\Delta = 1$, $\mu = 2.0861626e-07$, $\sigma = 1$

Test Results for $\Delta = 1.0000$

Number of degrees of freedom : 8
Value of the statistic : 19.15
p-value of test : 0.01

CPU time used : 00:00:00.00

Generator state:
486539008 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 114.15
p-value of test : 0.80

CPU time used : 00:00:00.01

Generator state:

503316224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33526.15
p-value of test : 1.6e-3

CPU time used : 00:00:00.03

Generator state:
520093440 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 12672.86
p-value of test : eps *****

CPU time used : 00:00:00.00

Generator state:
536870656 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 5258.52
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:

553647872 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.66

p-value of test : 0.26

CPU time used : 00:00:00.00

Generator state:

570425088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32

ChiSquare statistic : 3037.00

p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 31

ChiSquare statistic : 1477.92

p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom	:	32	
ChiSquare statistic	:	476.89	
p-value of test	:	eps	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	25	
ChiSquare statistic	:	29.81	
p-value of test	:	0.23	

Test on the values of the Statistic C

Number of degrees of freedom	:	15	
ChiSquare statistic	:	19.60	
p-value of test	:	0.19	

CPU time used : 00:00:00.07

Generator state:
587202304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	1173.47	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	629.20	
p-value of test	:	eps	*****

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	1.56
p-value of test	:	0.46

CPU time used : 00:00:00.01

Generator state:

620756480 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	8
Value of the statistic	:	5.79
p-value of test	:	0.67

CPU time used : 00:00:00.00

Generator state:
637533696 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	128
Value of the statistic	:	111.89
p-value of test	:	0.84

CPU time used : 00:00:00.01

Generator state:
654310912 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

```

-----
N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
Delta = 1, Mu = 0.00097653271, Sigma = 1

```

```

-----
Test Results for Delta = 1.0000

```

```

Number of degrees of freedom : 32768
Value of the statistic : 32572.20
p-value of test : 0.78

```

```

-----
CPU time used : 00:00:00.03

```

```

Generator state:
671088128 bits have been read.

```

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

```

```

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

```

```

sstring_HammingIndep test:

```

```

-----
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

```

```

Counters with expected numbers >= 10

```

```

-----
Number of degrees of freedom : 177
Chi-square statistic : 162.88
p-value of test : 0.77

```

```

-----
CPU time used : 00:00:00.00

```

```

Generator state:
687865344 bits have been read.

```

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

```

```

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

```

unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 314.24
p-value of test : 0.35

CPU time used : 00:00:00.01

Generator state:

704642560 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.78
p-value of test : 0.78

CPU time used : 00:00:00.01

Generator state:

721419776 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	26.49
p-value of test	:	0.74

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	32.44
p-value of test	:	0.40

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	31.26
p-value of test	:	0.50

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	20.31
p-value of test	:	0.73

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	29.64
p-value of test	:	0.01

CPU time used : 00:00:00.05

Generator state:
738196992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	2016.40	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	858.53	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	219.73	
p-value of test	:	1.2e-3	

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	38.93	
p-value of test	:	0.97	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	25.21	
p-value of test	:	0.80	

CPU time used : 00:00:00.04

Generator state:
754973952 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.24
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
9 RandomWalk1 H (L = 320)	eps

9 RandomWalk1 M (L = 320) eps

All other tests were passed

XX

Starting Alphabet: nb = 16777216

Version: TestU01 1.2.3

XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216

Number of cells = 2^L = 4

Expected number per cell = 4194304

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	5.50
p-value of test	:	0.06

CPU time used : 00:00:00.03

Generator state:

771751168 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE


```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic       : 5.06
p-value of test              : 0.75
```

CPU time used : 00:00:00.03

Generator state:
788528384 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32
```

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic       : 110.15
p-value of test              : 0.87
```

CPU time used : 00:00:00.00

Generator state:
805305600 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32705.50
p-value of test : 0.60

CPU time used : 00:00:00.03

Generator state:
822082816 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 174.33
p-value of test : 0.54

CPU time used : 00:00:00.00

Generator state:

838860032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	251.84
p-value of test	:	0.99

CPU time used	:	00:00:00.01
---------------	---	-------------

Generator state:

855637248 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	-1.13
p-value of test	:	0.87

CPU time used	:	00:00:00.00
---------------	---	-------------

Generator state:

872414464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32
```

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	20.75
p-value of test	:	0.94

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	27.36
p-value of test	:	0.65

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	15.28
p-value of test	:	0.9946

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	23.56
p-value of test	:	0.54

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	8.46
p-value of test	:	0.90

CPU time used : 00:00:00.08

Generator state:
889191680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	1902.23	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	914.68	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	256.49	
p-value of test	:	2.0e-6	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	52.79	
p-value of test	:	0.63	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	33.75	
p-value of test	:	0.38	

CPU time used : 00:00:00.05

Generator state:
905968640 bits have been read.

===== Summary results of Alphabit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin, unif01_CreateBitBlockGen:

Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.26
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
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9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	2.0e-6
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All other tests were passed